Born-Valid Dissipative Self-Interacting Dark Matter and the Compact Lens JVAS B1938+666

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Abstract. We investigate whether the compact strong-lens perturber in JVAS B1938+666 provides evidence for dissipative self-interacting dark matter (dSIDM) in controlled Born-valid particle models. Velocity-dependent elastic scattering and radiative energy loss are coupled self-consistently to gravothermal halo evolution and tested against the projected masses measured at 20 and 90 pc. Although the physical halo is slow, low-threshold branches explicitly open the radiative channel. Their exact cooling rates nevertheless remain negligible: the shortest cooling time is 8.7×10^{22} Gyr, and dissipative runs are indistinguishable from cooling-disabled controls. The unevolved NFW family can match both projected masses, but its isolated extrapolation is unusually concentrated. Tidal-envelope and host-orbit analyses show that environmental interpretation depends strongly on halo priors and line-of-sight geometry. An independent GM068 visibility audit and source-marginalized likelihood framework provide a consistency check on the lensing observables used in the gravothermal analysis. We find no positive evidence for radiative self-interactions in the explored Born-valid channels, including radiatively open thresholds. This result constrains the interpretation of B1938+666 within these microscopic models but does not exclude other dSIDM regimes.

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1. Introduction

The standard collisionless ΛCDM model reproduces the large-scale matter distribution, but galaxy-scale observations motivate tests of its assumptions about dark-matter microphysics. The central-density, diversity, and too-big-to-fail discussions are not each a unique failure of ΛCDM, but together they motivate models in which dark matter can exchange momentum and heat (Bullock & Boylan-Kolchin, 2017). Self-interacting dark matter (SIDM) is the minimal realization of this idea: elastic scattering redistributes energy inside a halo without introducing a new luminous component. Early cosmological calculations established the resulting core formation and changes to halo structure (Spergel & Steinhardt, 2000; Davé et al., 2001; Colín et al., 2002), while later simulations showed how the effect depends on halo mass, concentration, and the required cross section (Elbert et al., 2015; Garrison-Kimmel et al., 2013; Tulin & Yu, 2018).

The astrophysical interpretation is necessarily velocity dependent. A constant cross section per unit mass is a useful benchmark, but light-mediator models naturally produce transfer cross sections that vary strongly with relative velocity (Tulin et al., 2013; Kaplinghat et al., 2016). Resonant and long-range regimes can therefore soften dwarf-galaxy tensions while avoiding the much tighter limits from clusters. The Bullet Cluster remains a direct high-velocity test (Randall et al., 2008), and cluster strong lensing and group kinematics provide complementary constraints on the same velocity-dependent interaction (Andrade et al., 2022; Kahlhoefer et al., 2014; Sagunski et al., 2021). Baryons and the angular structure of scattering can also alter the mapping from a microscopic cross section to a halo profile (Robertson et al., 2017; Shen et al., 2021). These issues are now sufficiently developed that SIDM is best treated as a constrained particle-astrophysics framework rather than as a single constant-σ/m model (Adhikari et al., 2025).

Elastic scattering alone already leads to gravothermal evolution. Heat flows outward from a hot center, a core can form transiently, and the subsequent loss of pressure support can drive contraction or core collapse (Balberg et al., 2002; Koda & Shapiro, 2011; Nishikawa et al., 2020; Outmezguine et al., 2023). If a scattering event also radiates energy into a dark-sector state, the cooling channel competes with conductive transport and can accelerate this sequence. Dissipative dark-sector constructions have been studied in the context of dark disks, bound states, and compact-object formation (Fan et al., 2013; Wise & Zhang, 2014; Essig et al., 2019; Xiao et al., 2021). A phenomenological cooling coefficient is useful for isolating this dynamical mechanism, but it does not guarantee that a particle model can realize the assumed cooling rate at the velocities present in the halo.

Strong gravitational lensing supplies an unusually clean way to test this question. A small perturber is detected through its gravitational potential, so the inference does not require a luminous counterpart, and gravitational imaging can constrain projected mass on tens-of-parsecs scales. This is the role of the compact perturber in JVAS B1938+666: a roughly million-solar-mass object at the lens redshift $z = 0.881$ was identified through its effect on the lensed surface brightness (Powell et al., 2025; Vegetti et al., 2026). The inferred profile is summarized particularly efficiently by two projected masses, inside 20 and 90 pc, which constrain both the central normalization and the profile shape. It is therefore a useful case study for gravothermal models, not an assumption that the perturber has already been proven to be dSIDM.

Using velocity-independent transfer and dissipation parameters, Schmidt et al. (2026) showed that both elastic and weakly dissipative SIDM halos can reproduce these two masses near the late stages of gravothermal evolution. Their phenomenological result establishes a conditional possibility: dissipation can reduce the evolution time or, equivalently, the cross section required to obtain a compact halo. The remaining question is whether the same conclusion survives when the cooling and scattering rates are tied to a controlled microscopic model.

A light mediator produces a velocity-dependent transfer cross section, while emission of a massive boson introduces a threshold in relative velocity. These scales break the rescaling symmetry of a constant-cross-section halo and can suppress dissipation in low-velocity systems. In addition, perturbative cross sections are predictive only inside their domain of validity. The relevant question is therefore not whether a prescribed constant cooling parameter can affect a halo, but whether a Born-valid microscopic model populates the radiative region after the B1938 halo has been placed in physical units.

We address this question with three ingredients. First, we implement the Born-limit elastic transfer cross section and the long-range identical-fermion emission kernels derived by Lankester-Broche & Pradler (2026). Second, we perform an explicit Born-valid parameter scan rather than assigning an independent cooling strength. Third, we use the constant-model rescaling only to propose physical initial conditions and then rerun every candidate directly, preserving its microscopic velocity scale. The result is a self-consistent test of whether B1938 probes dissipation in this class of models.

This paper is organized as follows. Section 2 describes the scattering models and cooling moment. Section 3 summarizes the fluid solver, B1938 statistic, and scan strategy. Section 4 presents the Born-valid parameter map, direct 192-shell time scan, physical threshold audit, elastic controls, a dimensionless cooling-failure map, a cosmological-profile audit, and the GM068 visibility-likelihood and host-orbit sensitivity analyses. We discuss the physical interpretation and limitations in Section 5 and conclude in Section 6.

2. Microphysical framework

2.1. Velocity-dependent scattering and Born validity

For identical particles of mass m_χ and reduced mass $\mu = m_\chi/2$, we use the Born-limit transfer cross section

$$\sigma_T(v_{\rm rel}) = \frac{a^4}{8\pi\mu^2 v_{\rm rel}^4}\left[\ln\left(1+\frac{4\mu^2 v_{\rm rel}^2}{m_{\rm med}^2}\right) - \frac{4\mu^2 v_{\rm rel}^2/m_{\rm med}^2}{1+4\mu^2 v_{\rm rel}^2/m_{\rm med}^2}\right], \tag{1}$$

where $m_{\rm med}$ is the mediator mass and a denotes the effective coupling entering the scattering amplitude. The cross section approaches a constant at small velocity and falls approximately as $v_{\rm rel}^{-4}$, up to a logarithm, at large velocity.

The perturbative expansion is monitored with

$$\eta_B = \frac{\alpha_D \mu}{m_{\rm med}}. \tag{2}$$

We classify $\eta_B < 1$ as Born-valid and use $\eta_B = 0.1$ for the final controlled candidates. The emission threshold is parameterized by

$$v_\star = \sqrt{\frac{2m_{\rm med}}{\mu}}. \tag{3}$$

For a chosen $(\eta_B, v_\star)$, the dark-matter and mediator masses are varied together to obtain a target value of σ_T/m_χ at 100 $\mathrm{km\,s^{-1}}$.

We study two emission channels. Model M1 contains massive-vector emission in long-range χ–V scattering, while M2 contains massive-scalar emission in long-range χ–ϕ scattering. The elastic parameters can be chosen to give the same reference cross section and threshold, but the two channels retain different energy-differential emission kernels. This construction separates differences caused by radiation from differences in the elastic normalization.

We do not include the legacy massless-emission model as a third predictive channel. A strictly massless long-range elastic mediator has an infrared-divergent Rutherford transfer cross section, so the earlier control required an arbitrary finite-mass regulator. Its constant dissipation normalization was also a provisional phenomenological closure rather than the validated energy moment used for M1 and M2. A physical massless calculation would require a specified screening scale and an inclusive treatment of soft radiation. It is therefore outside the controlled scope of this work.

2.2. Energy-weighted cooling moment

Let ω be the radiated energy in a collision. For each relative speed we compute

$$Q(v_{\rm rel}) = \int_{m_{\rm em}}^{\mu v_{\rm rel}^2/2} d\omega\, \omega \frac{d\sigma}{d\omega}, \tag{4}$$

using the long-range identical-fermion kernels of Lankester-Broche & Pradler (2026). The corresponding cooling combination is

$$\left(\frac{\sigma}{m_\chi}\right)_{\rm cool}(v_{\rm rel}) = \frac{2Q(v_{\rm rel})}{m_\chi^2 v_{\rm rel}^2}. \tag{5}$$

The volumetric energy-loss rate for one identical population is

$$C_{\rm vol} = \frac{\rho^2}{4}\left\langle \left(\frac{\sigma}{m_\chi}\right)_{\rm cool}(v_{\rm rel}) v_{\rm rel}^3 \right\rangle_{\rm rel}, \tag{6}$$

where the factor $1/4$ accounts for identical-particle pairs in the adopted normalization. The relative speed follows a Maxwell distribution whose component variance is twice the one-particle one-dimensional variance. This average includes the high-velocity tail and is therefore more informative than a sharp comparison between a single characteristic velocity and $v_\star$.

For constant σ_T/m_χ and a constant fractional dissipation parameter $r_{\rm diss}$, Eq. (6) reduces to

$$C_{\rm vol} = \frac{8}{\sqrt{\pi}}\frac{\sigma_T}{m_\chi}(r_{\rm diss}-1)\rho^2\nu^3, \tag{7}$$

where ν is the one-dimensional velocity dispersion. This is the constant-parameter limit used by Schmidt et al. (2026). In our microphysical model, the energy moment fixes the dissipative loss rather than an independent constant parameter.

3. Numerical method

3.1. Gravothermal evolution

We evolve spherical, isolated Navarro–Frenk–White (NFW) halos (Navarro et al., 1997) with the GravothermalSIDM fluid equations (Balberg et al., 2002; Koda & Shapiro, 2011; Nishikawa et al., 2020; Outmezguine et al., 2023; Gad-Nasr et al., 2024). The initial profile has scale radius $r_s = 3.6$ kpc and scale density $\rho_s = 7.09\times10^{-3}\,M_\odot\,\mathrm{pc}^{-3}$. The solver follows Lagrangian mass shells in hydrostatic equilibrium and interpolates between long- and short-mean-free-path heat conduction. At each shell, the elastic transport coefficient is thermally averaged from Eq. (1) and the specific cooling rate is $C_{\rm vol}/\rho$ from Eq. (6). The timestep limits both conductive and radiative changes in the specific internal energy.

The calculation is performed with 96 shells for parameter screening and 192 shells for the final candidates. A 48-shell grid was also tested, but it can shift the preferred snapshot and produce spuriously small mass residuals; it is therefore used only for software-level checks. The final timestep tolerance is $\epsilon_t = 0.005$.

3.2. B1938 projected masses

We adopt the B1938 projected masses

$$M_{20}^{\rm obs} = (4.25\pm0.21)\times10^5\,M_\odot, \tag{8}$$

$$M_{90}^{\rm obs} = (1.167\pm0.039)\times10^6\,M_\odot, \tag{9}$$

at projected radii of 20 and 90 pc (Vegetti et al., 2026). The corresponding observed ratio is 0.364 ± 0.022. Projected masses are calculated by integrating the exact cylinder-intersection volume of each piecewise-constant fluid shell.

For every physical direct run we use the two-mass statistic

$$\chi^2_{2M} = \left(\frac{M_{20}-M_{20}^{\rm obs}}{\delta M_{20}}\right)^2 + \left(\frac{M_{90}-M_{90}^{\rm obs}}{\delta M_{90}}\right)^2. \tag{10}$$

This statistic is a compact diagnostic based on two measurements; it is not a replacement for the full lens-imaging likelihood.

3.3. GM068 visibility audit and likelihood scope

We independently audit the pipeline-calibrated GM068 UVFITS data (Powell et al., 2025). The file contains 2,716,193 visibility groups, eight 8-MHz intermediate-frequency windows, and the parallel-hand correlations RR and LL. Of the 43,459,088 stored complex correlation samples, 30,173,833 have positive finite pipeline weights. Following the published procedure, we estimate one Gaussian noise standard deviation per real or imaginary component by differencing time-adjacent visibilities and accumulating the root-mean-square in 30-minute baseline bins. We remove the full EF–JB–WB triangle and retain the published RFI-sensitive baselines as a separate mask category. Two LA–PT bins at 9–10 h are additionally identified by a declared six-MAD threshold in log noise and included in the data-driven RFI mask.

The visibility file was averaged by AIPS `SPLIT` from 32 channels to one channel per 8-MHz window. Our visibility operator therefore averages the sky model over the 32 original channel centers. The independent forward model contains the published elliptical power-law macromodel, $m = 3$ and 4 convergence multipoles, external shear, spherical pseudo-Jaffe or tabulated perturbers, a bilinear pixelated source, and forward/adjoint non-uniform fast Fourier transforms. For finite test problems we evaluate the normalized Gaussian source-marginalized likelihood, including the precision-matrix determinants.

This implementation establishes data provenance and validates the visibility bookkeeping, noise model, bandwidth integration, forward/adjoint operators, and source-marginalized likelihood. The gravothermal calculation retains Eq. (9) as its observational compression, while the visibility analysis supplies an independent consistency check on the data and likelihood construction.

At each source snapshot, a logarithmic r_{2D}/r_s grid first locates the relevant rescaling branch, after which a bounded continuous minimization of Eq. (10) determines the radius scale. The common mass normalization is solved analytically. This refinement is essential: choosing only among discrete radius samples produces artificial jumps when adjacent grid points exchange rank. It also makes explicit that two continuous NFW rescaling parameters can interpolate two projected masses at $t = 0$ essentially exactly. The diagnostic power therefore comes from direct evolution and from physical priors, not from the zero-time residual alone.

3.4. Direct physical resimulation

For constant cross sections, the transformation

$$r \to \lambda r, \qquad M \to \mu M, \qquad t \to \left(\frac{\lambda^3}{\mu}\right)^{1/2} t, \qquad \frac{\sigma}{m} \to \frac{\lambda^2}{\mu}\frac{\sigma}{m} \tag{11}$$

maps one gravothermal solution into another (Schmidt et al., 2026). A massive emission threshold is not invariant under this transformation because the velocity changes as $(\mu/\lambda)^{1/2}$. We therefore use Eq. (11) only to infer a candidate physical r_s, ρ_s, and target time. A new 192-shell halo is then evolved with those physical parameters and the unchanged microscopic model. The final masses in Eq. (10) are measured from this direct run, without a post-evolution mass rescaling.

The source-snapshot grid for every final run is

$$t_{\rm source}/{\rm Gyr} = 0, 0.01, 0.02, 0.03, 0.04, 0.05, 0.10. \tag{12}$$

The explicit zero-time point identifies the interpolating initial condition. The physical target time of a direct run is reported separately from $t_{\rm source}$.

3.5. Physical activity audit and controls

For each direct halo we compare the maximum physical one-dimensional dispersion with the emission threshold and evaluate the shortest local cooling time,

$$t_{\rm cool}^{\rm min} = t_{\rm scale} \min_i \left(\frac{u_i}{C_{{\rm cool},i}} \right), \tag{13}$$

where u_i and $C_{{\rm cool},i}$ are in code units. The exact thermal kernel remains in Eq. (13), so a possible Maxwell tail above threshold is retained. All 12 candidates in the radiatively open $v_\star = 1$ and 5 $\rm km\,s^{-1}$ branches are rerun at $t_{\rm source} = 0.10$ Gyr with cooling disabled but identical velocity-dependent elastic transport.

3.6. Dimensionless failure map and halo-prior audit

To express the cooling hierarchy independently of a chosen evolution time, we compare Eq. (13) with the local dynamical time at the same fastest-cooling shell,

$$t_{\rm dyn}(r_i) = \left[4\pi G \rho(r_i)\right]^{-1/2}. \tag{14}$$

On the common unevolved 192-shell physical halo, we evaluate the exact thermal kernel on a 49-point grid in $0.1 \leq v_\star/v_{\rm max} \leq 100$. At fixed threshold and $\eta_B = 0.1$, the source term scales linearly with the reference cross section, allowing a 41-point grid over $0.1 \leq \sigma_T/m_\chi(100) \leq 10\,{\rm cm}^2\,{\rm g}^{-1}$. The resulting 4018 points are an initial-state post-processing diagnostic, not additional halo evolutions. Existing direct candidates provide an independent overlay.

We separately test whether the zero-time NFW interpolation resembles a cosmological halo. The median Planck concentration relation is

$$\log_{10} c_{200} = a(z) + b(z) \log_{10} \left[\frac{M_{200}}{10^{12} h^{-1} M_\odot} \right], \tag{15}$$

where $a = 0.520 + (0.905 - 0.520)\exp(-0.617 z^{1.21})$ and $b = -0.101 + 0.026z$ (Dutton & Macciò, 2014); we adopt its 0.11-dex intrinsic scatter. To allow a broad tidal envelope, we forward-project the smoothly truncated profile

$$\rho_{\rm BMO}(r) = \rho_{\rm NFW}(r) \left(\frac{r_t^2}{r^2 + r_t^2} \right)^2 \tag{16}$$

(Baltz et al., 2009) and profile the two masses over $10^{5.5} \leq M_{200}/M_\odot \leq 10^{10}$ and $1 \leq r_t/r_s \leq 100$. A deliberately aggressive sensitivity branch uses $r_t/r_s \geq 0.5$. We quote the concentration offset at which the profiled two-mass statistic reaches $\chi^2_{2M} = 2.30$ only as a common reference contour. With two data points and profiled halo/truncation parameters, it is not a posterior credible interval or a detection significance.

3.7. Host-orbit and tidal sensitivity prior

The preceding BMO envelope treats r_t/r_s as a free sensitivity coordinate. We separately propagate the unknown three-dimensional location through the tidal prescription used in the published lens analysis. The tidal pseudo-Jaffe fit gives $r_{t,0} = 53 \pm 1$ pc for $m_0 = (1.54 \pm 0.07) \times 10^6\, M_\odot$ when the true host distance is set equal to the projected distance (Powell et al., 2025). The displayed macromodel and perturber posterior medians give $R_{\rm proj} \simeq 1.52$ kpc and $\gamma \simeq 1.76$. Since $M_{\rm host}(< R) \propto R^{3-\gamma}$ for the power-law host, the same prescription scales as

$$r_t(r_p, m) = 53 \text{ pc} \left(\frac{m}{1.54 \times 10^6\, M_\odot}\right)^{1/3} \left(\frac{r_p}{1.52 \text{ kpc}}\right)^{\gamma/3}, \tag{17}$$

where r_p is the orbital pericenter.

We write the current radius as $r = [R^2_{\rm proj} + z^2_{\rm los}]^{1/2}$ and $r_p = f_p r$. The conditional line-of-sight sensitivity prior follows an NFW-like tracer number density with a 30-kpc scale radius and a 200-kpc outer boundary. We compare a circular upper envelope $f_p = 1$, a phase-mixed family $f_p \sim \mathrm{Beta}(4, 2)$, and a radially biased sensitivity family $f_p \sim \mathrm{Beta}(2, 4)$. The free pseudo-Jaffe summaries $m = (2.82 \pm 0.26) \times 10^6\, M_\odot$ and $r_t = 149 \pm 18$ pc supply the likelihood. They are treated as independent Gaussians in this sensitivity calculation. These choices define an explicit robustness audit, so the resulting distributions quantify prior dependence rather than a calibrated cosmological orbit posterior.

The numerical implementation is covered by 48 regression tests. They check the relative-speed distribution, the constant-parameter limit of Eq. (6), the massless analytic emission rate, threshold closure of the massive kernels, Born-parameter construction, exact shell projection, continuous candidate refinement, direct-snapshot selection, and the cooling/dynamical-time and cosmological-profile identities. The added visibility tests cover AIPS baseline decoding, RR/LL flag handling, bandwidth integration, complex-likelihood normalization, pseudo-Jaffe mass normalization, the NUFFT adjoint identity, bilinear source mapping, and exact finite-dimensional source marginalization.

4. Results

4.1. Born-valid parameter space

The Born condition substantially restricts a direct interpretation of large phenomenological dSIDM cross sections. If only the coupling is varied at the nominal M1 and M2 mass hierarchies, the inferred expansion parameters are much larger than unity. At the limiting value $\eta_B = 1$, the largest transfer cross sections at 100 $\mathrm{km\,s^{-1}}$ are approximately 3.23×10^{-4} and $9.72 \times 10^{-3}\,\mathrm{cm^2\,g^{-1}}$ for M1 and M2, respectively. Thus a fixed

Table 1. Numerical scans used in the analysis. Cross sections are in $\mathrm{cm^2\,g^{-1}}$ and velocities in $\mathrm{km\,s^{-1}}$. Every final point is evaluated for both models and seven source times.

Stage	shells	η_B	$\sigma_T/m_\chi(100)$	$v_\star$
Born screening	96	$0.03, 0.1, 0.3$	$0.1, 1, 10$	100
Open/near-threshold final grid	192	0.1	$0.1, 1, 10$	$1, 5, 20$
High-threshold final controls	192	0.1	0.1	$100, 500$
Cooling-disabled controls	192	0.1	$0.1, 1, 10$	$1, 5$

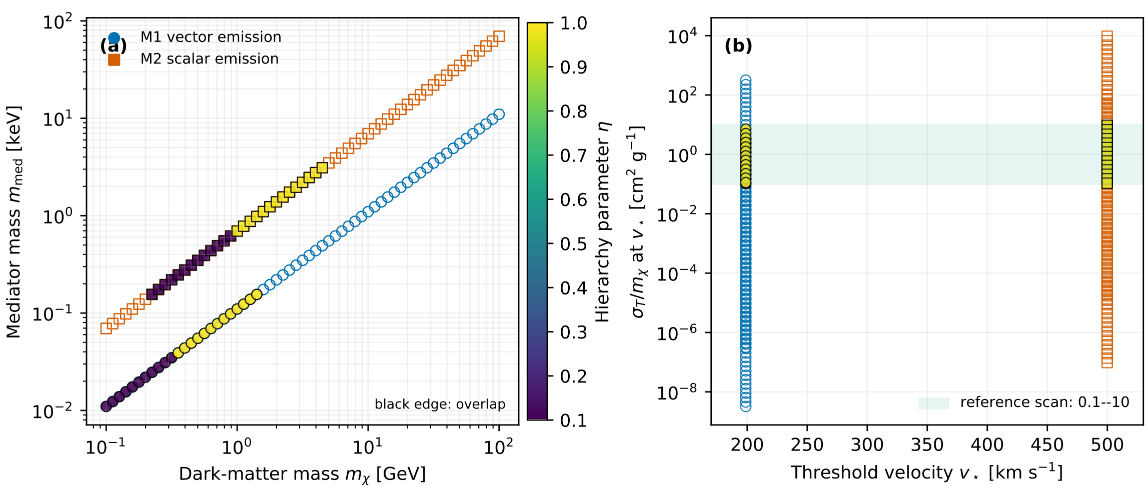

Figure 1. Born-valid mass-hierarchy screening. (a) Dark-matter and mediator masses in the hierarchy scan; colors encode η_B, and black-edged points satisfy the astrophysical overlap criterion. (b) Transfer cross section per mass at the imposed threshold velocity. The shaded band marks the 0.1–10 $\mathrm{cm^2\,g^{-1}}$ reference range used for the controlled scan.

normalization of order $50\,\mathrm{cm^2\,g^{-1}}$ cannot be combined with these nominal masses in the Born regime.

A Born-valid astrophysical window opens when the mass hierarchy is varied at fixed threshold. For $\eta_B = 0.1$ and $0.1 \leq \sigma_T/m_\chi(100\,\mathrm{km\,s^{-1}}) \leq 10\,\mathrm{cm^2\,g^{-1}}$, the corresponding dark-matter masses are approximately 0.10–0.32 GeV for M1 and 0.22–0.89 GeV for M2 at their nominal thresholds. This motivates a controlled scan in which η_B, the reference cross section, and $v_\star$ are treated as explicit coordinates.

Figure 1 visualizes the mass-hierarchy scan behind this selection. The filled points satisfy the astrophysical overlap criterion, whereas open points are Born-valid but outside that reference window. The direct calculations therefore sample a controlled subset of the broader microphysical construction rather than treating every Born-valid point as an equally plausible B1938 candidate.

Table 2. Particle-mass ranges in the $\eta_B = 0.1$ final grid. Each range runs from $\sigma_T/m_\chi(100) = 10$ to $0.1\,\mathrm{cm^2\,g^{-1}}$.

$v_\star$ [km s^{-1}]	m_χ [MeV]	m_{med} [keV]
1	0.0792–0.368	2.20×10^{-10}–1.02×10^{-9}
5	0.634–2.94	4.41×10^{-8}–2.05×10^{-7}
20	3.76–17.5	4.18×10^{-6}–1.94×10^{-5}

4.2. Screening and final-grid construction

The 96-shell $v_\star = 100\,\mathrm{km\,s^{-1}}$ scan contains 18 completed fluid evolutions spanning the screening coordinates in Table 1. Its restored physical halos have $v_{\max}$ between 5.23 and 5.38 $\mathrm{km\,s^{-1}}$, and the emission kernel is numerically zero. Consequently, variations across that branch measure elastic transport and not radiation. The final calculation fixes the explicitly controlled value $\eta_B = 0.1$, retains all three reference cross sections at $v_\star = 1, 5, 20\,\mathrm{km\,s^{-1}}$, and uses the 100 and 500 $\mathrm{km\,s^{-1}}$ points as high-threshold controls.

Table 2 gives the particle scales of the new low-threshold grid. Lowering $v_\star$ while fixing η_B and the reference cross section changes m_χ, m_{med}, and the coupling together; it does not merely remove a step function from an otherwise fixed model.

4.3. Uniform 192-shell time scan

The final grid contains 22 parameter/model combinations and seven source times, for 154 completed direct runs. M1 and M2 agree exactly in the projected masses at the saved precision. Table 3 reports the common χ^2_{2M} at the last source snapshot. At fixed threshold, increasing the reference cross section accelerates elastic evolution, but none of the curves develops an interior minimum.

Figure 2 shows the full threshold comparison at $\sigma_T/m_\chi(100) = 0.1\,\mathrm{cm^2\,g^{-1}}$ and the cross-section comparison at $v_\star = 5\,\mathrm{km\,s^{-1}}$. The continuous source-profile optimization removes the grid-switching artifacts present in a purely discrete radius selection. Every curve is minimized at the initial state, where

$$\chi^2_{2M} = 2.1 \times 10^{-16}, \qquad \frac{M_{20}}{M_{90}} = 0.3641817. \tag{18}$$

This numerical zero is expected because two NFW rescaling parameters are being fitted to two masses. It demonstrates interpolation within the adopted initial family, not preference for an interaction history.

The nonzero source times map to much shorter physical times. At the largest source time in Eq. (12), the direct target time is approximately 1.1–1.2 Myr. The increase of χ^2_{2M} is controlled by elastic evolution: it depends strongly on the reference cross section, while M1 and M2 remain coincident.

Table 3. Direct two-mass statistic at $t_{\rm source} = 0.10$ Gyr in the final 192-shell scan. M1 and M2 are identical at the reported precision.

$v_\star$ [km s^{-1}]	$\sigma_T/m_\chi(100)$ [cm^2 g^{-1}]	χ^2_{2M}
1	0.1	5.47×10^{-5}
1	1	7.21×10^{-3}
1	10	6.20×10^{-2}
5	0.1	1.98×10^{-5}
5	1	6.21×10^{-3}
5	10	6.40×10^{-2}
20	0.1	1.48×10^{-6}
20	1	4.91×10^{-3}
20	10	6.36×10^{-2}
100	0.1	1.50×10^{-4}
500	0.1	2.30×10^{-3}

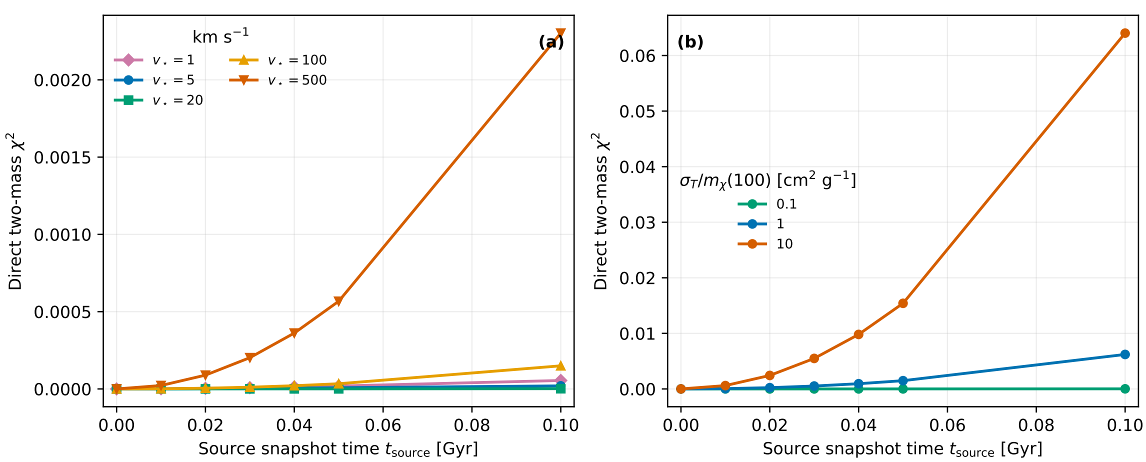

Figure 2. Extended 192-shell direct time scan. M1 and M2 overlap at the plotted precision. (a) Threshold comparison at $\sigma_T/m_\chi(100) = 0.1\,{\rm cm^2\,g^{-1}}$. (b) Reference-cross-section comparison at $v_\star = 5\,{\rm km\,s^{-1}}$. The smooth curves use the continuous source-profile optimization described in Section 3.

4.4. Physical threshold and cooling audit

The physical velocity hierarchy is summarized in Table 4. The maximum direct dispersion is approximately $5.34\,{\rm km\,s^{-1}}$ for every candidate. The 1 and 5 km s^{-1} branches therefore reach the massive-emission region, whereas the 20, 100, and 500 km s^{-1} branches remain below threshold. Figure 3 separates this reachability test from the cooling-time calculation.

The lowest cooling time occurs for M1 at $v_\star = 5\,{\rm km\,s^{-1}}$ and $\sigma_T/m_\chi(100) = 10\,{\rm cm^2\,g^{-1}}$. It is still 8.69×10^{22} Gyr. At fixed threshold the cooling time decreases approximately in inverse proportion to the reference cross section, but the factor of 100

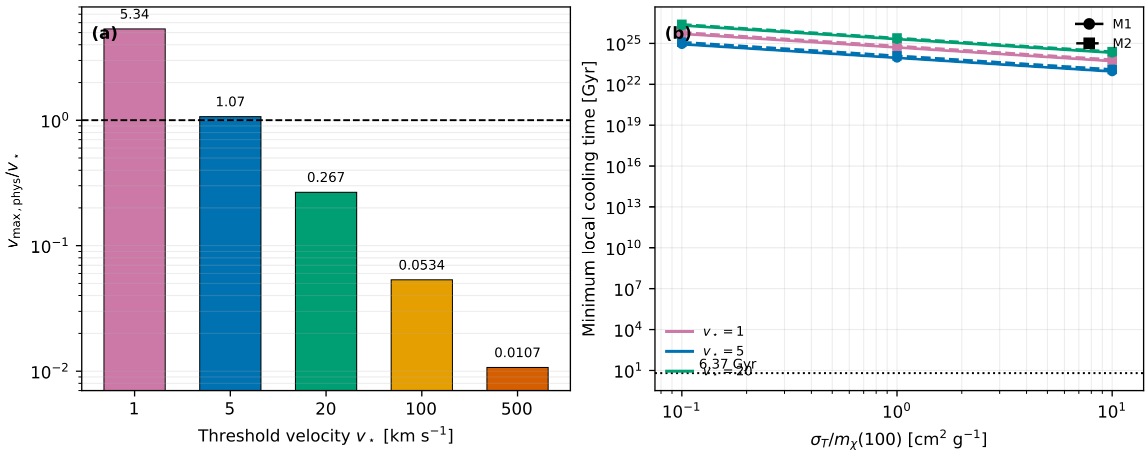

Figure 3. Physical activity audit. (a) Maximum physical one-dimensional dispersion divided by the emission threshold; the dashed line marks unity. (b) Minimum exact local cooling time for the finite-kernel branches. Colors denote thresholds and line styles denote emission channels. The dotted line marks the 6.37 Gyr cosmic age at the lens redshift.

Table 4. Physical activity audit of the final candidates. The cooling-time column gives the minimum over both models and all tested reference cross sections; an infinite value denotes a numerically vanishing thermal kernel.

$v_\star$ [km s^{-1}]	$v_{\rm max}$ [km s^{-1}]	$v_{\rm max}/v_\star$	minimum $t_{\rm cool}^{\rm min}$ [Gyr]
1	5.344	5.34	5.02×10^{23}
5	5.344	1.069	8.69×10^{22}
20	5.344	0.267	2.06×10^{24}
100	5.344	0.0534	∞
500	5.349	0.0107	∞

across the scan is negligible compared with the more than 20-order-of-magnitude gap to a cosmological time. The 1 km s^{-1} cooling time is longer than the 5 km s^{-1} value because the fixed-$(\eta_B, \sigma_T/m_\chi)$ construction changes the masses and coupling along with the threshold. Threshold opening alone therefore does not determine the cooling hierarchy.

The cooling-disabled controls provide a direct test of the elastic limit. At $t_{\rm source} =$ 0.10 Gyr, all 12 controls in the open-threshold branches use the same physical initial conditions, elastic transport, resolution, and timestep tolerance as their dissipative counterparts. For every pair we find

$$\Delta\chi^2_{2M} = 0, \qquad \Delta(M_{20}/M_{90}) = 0 \tag{19}$$

at the saved precision, with identical conduction-step counts. The equality in Eq. (19) verifies that even the radiatively open candidates are in the effective elastic limit.

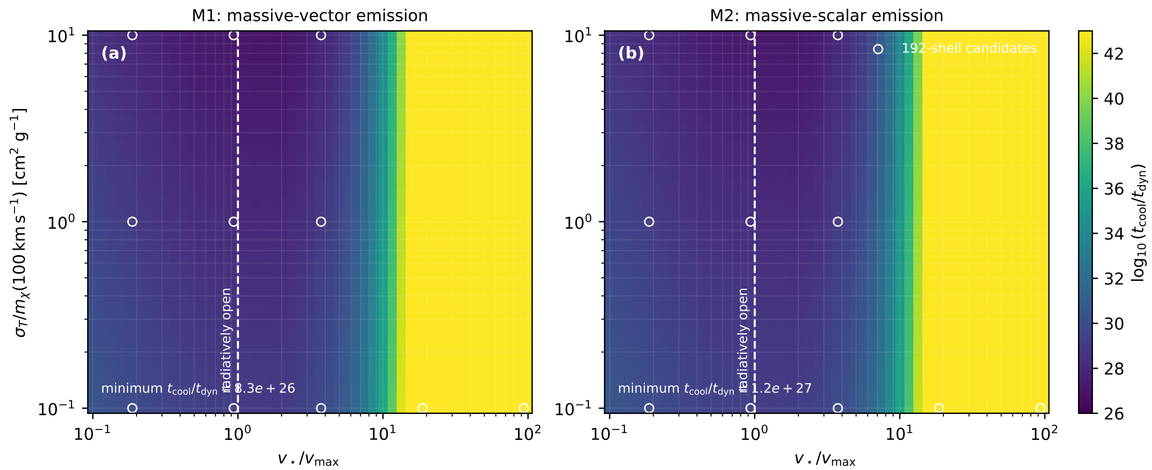

Figure 4. Born-controlled initial-state cooling-failure map for (a) massive vector and (b) massive scalar emission. The color gives the local ratio $t_{\rm cool}/t_{\rm dyn}$, evaluated at the shell with the shortest cooling time; the vertical dashed line marks $v_\star = v_{\rm max}$. Open circles are the direct 192-shell candidates. The dense grid is a kernel post-processing diagnostic, not an additional time-evolution scan.

4.5. Dimensionless cooling-failure map

Figure 4 generalizes the finite direct grid in the two coordinates that control whether massive emission can matter. The shortest cooling times over the dense map are 8.09×10^{22} Gyr for M1 and 1.14×10^{23} Gyr for M2, both at $v_\star/v_{\rm max} = 0.75$ and $\sigma_T/m_\chi(100) = 10\,\mathrm{cm^2\,g^{-1}}$. Thus even the most favorable point cools more than 1.2×10^{22} times slower than the 6.37 Gyr age at the lens redshift.

The minimum dynamical margin is $t_{\rm cool}/t_{\rm dyn} = 8.32\times10^{26}$ for M1 and 1.18×10^{27} for M2, occurring near $v_\star/v_{\rm max} = 1.78$ at the largest cross section. The different locations of the minimum absolute cooling time and minimum local ratio reflect changes in the shell that minimizes Eq. (13). More importantly, neither threshold crossing nor the two-decade cross-section lever arm approaches the dynamically relevant region $t_{\rm cool}/t_{\rm dyn} \lesssim 1$.

4.6. Cosmological concentration and tidal-envelope audit

The continuously fitted initial profile has

$$r_s = 10.00\ \mathrm{pc}, \qquad \rho_s = 55.88\ M_\odot\,\mathrm{pc}^{-3}. \tag{20}$$

Treating this profile as an untruncated isolated NFW halo, we adopt $H_0 = 67.4$ km s^{-1} Mpc^{-1} and $\Omega_m = 0.315$. This gives

$$c_{200} = 219, \qquad r_{200} = 2.19\ \mathrm{kpc}, \qquad M_{200} = 3.09\times10^6\ M_\odot. \tag{21}$$

At this mass and $z = 0.881$, Eq. (15) gives $c_{200} = 15.50$. The isolated extrapolation is therefore 10.46 units of the assumed 0.11-dex log-concentration scatter above the median, rather than an ordinary $c \sim 15$–17 halo.

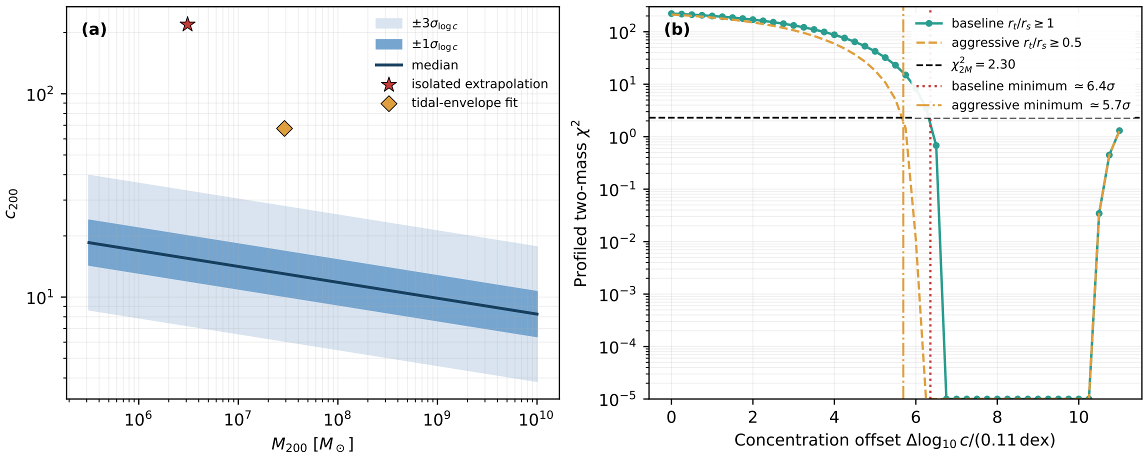

Figure 5. Cosmological and tidal-envelope plausibility audit. (a) The Dutton & Macciò (2014) median concentration relation at $z = 0.881$, with one- and three-scatter bands, compared with the isolated extrapolation and the first baseline tidal-envelope grid point inside the reference contour. (b) Two-mass χ^2 profiled over M_{200} and BMO truncation at fixed concentration offset. The baseline has $r_t/r_s \geq 1$; the aggressive sensitivity branch extends to 0.5. Offset labels are in units of the assumed 0.11-dex scatter and are not formal statistical significances.

When M_{200} and $1 \leq r_t/r_s \leq 100$ are profiled, the two-mass statistic first reaches the reference contour at an interpolated concentration offset of 6.36 scatter units. The first sampled solution inside the contour has $M_{200} = 2.91 \times 10^7\, M_\odot$, $c_{200} = 67.5$, and saturates the lower truncation bound $r_t/r_s = 1$. Extending that bound to the deliberately aggressive value 0.5 lowers the interpolated offset only to 5.69 units and again drives the fit to the boundary. The boundary saturation makes the dependence on the assumed tidal envelope explicit.

Equation (21) is an extrapolation diagnostic rather than a concentration measurement. The fluid model extends only to $50 r_s \simeq 0.50$ kpc, whereas the inferred r_{200} is $219 r_s$; moreover, a tidally stripped subhalo need not retain a meaningful isolated-halo c_{200}. The zero-time mass match therefore shows only that a freely rescaled NFW shape can interpolate the two measurements. The audit in Figure 5 shows that neither an ordinary untruncated halo nor the adopted broad truncation envelope makes this particular interpolation typical. This BMO profile exercise alone does not provide a formal exclusion of a CDM subhalo because its truncation coordinate is not tied to an infall history or a full lens likelihood.

4.7. GM068 likelihood audit and host-orbit sensitivity

The visibility audit yields 2255 baseline/time noise bins. Of these, 76 lie in the published EF–JB–WB triangle, 16 contain too few adjacent pairs for the adopted noise estimate, and two LA–PT bins are selected by the declared data-driven RFI rule. After polarization averaging and these exclusions, 15,923,560 Stokes-I samples enter the full-

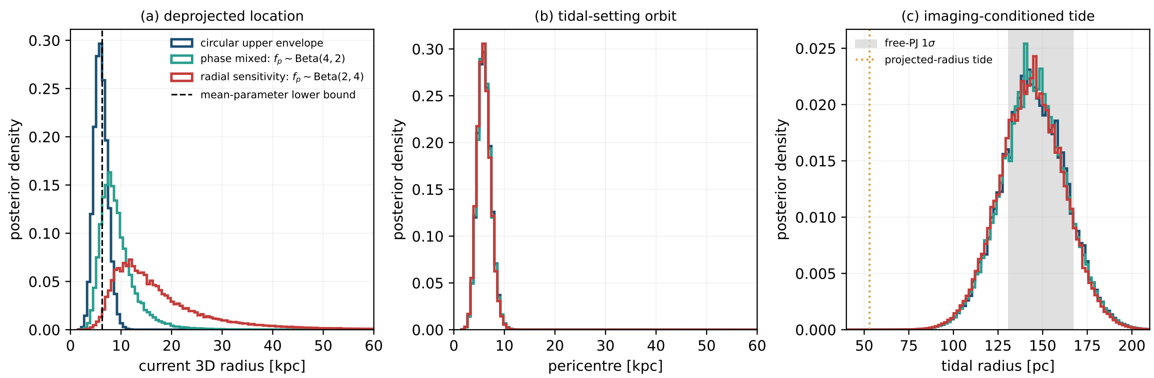

Figure 6. Host-orbit and tidal sensitivity audit conditioned on the published free pseudo-Jaffe summaries. (a) Current three-dimensional radius for the circular upper envelope, phase-mixed prior, and radially biased sensitivity prior. The dashed line is the lower bound evaluated at the mean measured mass and truncation radius; measurement uncertainty permits posterior samples below it. (b) Pericenter distributions that set the tidal radius. (c) Imaging-conditioned tidal-radius distributions; the gray band is the published free-PJ 1σ interval and the dotted line is the tidal radius obtained by setting the three-dimensional radius equal to the projected one. The curves propagate the published one-dimensional pseudo-Jaffe summaries under the declared orbital priors and represent sensitivity distributions rather than formal orbit constraints.

data dirty-image QA. The visibility forward/adjoint identity, band integration, and finite-dimensional source marginalization pass the regression suite. The QA also shows that the PRONTO image-plane coordinates are offset from the GM068 UVFITS phase center; a future independent fit must sample this global shift. These checks validate the visibility-likelihood pathway and support the use of Eq. (9) as the observational compression in the gravothermal calculation.

Equation (17) gives a geometric bound within the adopted tidal model: since $r_p \leq r$, the mean free-PJ mass and truncation radius require $r \geq 6.277$ kpc and $|z_{\rm los}| \geq 6.090$ kpc. This is substantially larger than $R_{\rm proj} \simeq 1.52$ kpc, quantifying why substituting projected distance for three-dimensional distance produces an overly compact tidal radius.

For the circular upper envelope, the current-radius posterior has median 5.97 kpc and 16–84 percentile interval 4.74–7.39 kpc. The phase-mixed family gives $8.57^{+3.68}_{-2.26}$ kpc, while the radially biased sensitivity family gives $14.97^{+11.26}_{-5.41}$ kpc. Their pericenter medians remain close to 5.9–6.0 kpc because the free-PJ truncation radius primarily constrains the tidal-setting pericenter. Changing the NFW-like tracer scale radius from 10 to 100 kpc moves the phase-mixed current-radius median only from 8.15 to 8.92 kpc. Orbital-family uncertainty therefore dominates this host-scale sensitivity. This analysis shows that line-of-sight geometry can reconcile the published projected-radius tide with a more extended perturber; it does not identify the perturber as CDM, SIDM, or dSIDM.

5. Discussion

5.1. From phenomenological cooling to microscopic thresholds

Velocity-independent dSIDM calculations establish an important conditional result: if the local cooling strength is appreciable, dissipation can change the gravothermal trajectory and accelerate the production of compact halos (Schmidt et al., 2026). Our calculation asks whether a specific perturbative emission model realizes that condition in the physical B1938 halo. It does not in the parameter region tested here.

This difference is not a disagreement about the fluid response to cooling. Equation (6) reproduces the constant-model source term in Eq. (7). The difference arises before the nonlinear halo response. At high thresholds the physical velocity distribution scarcely populates the massive-emission region. At $v_\star = 1$ and 5 $\mathrm{km\,s^{-1}}$ the region is populated, but the absolute emitted-energy moment remains negligible. The phenomenological and microscopic calculations therefore probe different cooling normalizations even when their kinematic regimes overlap.

The Born audit is central to this interpretation. A large independent value of $r_{\rm diss} - 1$ can be assigned in a model-agnostic fluid calculation, but the scattering and emission rates are linked once masses and couplings are specified. At fixed nominal mass hierarchy, the large cross sections often used to produce rapid gravothermal evolution lie outside the Born expansion. Moving to a Born-valid hierarchy can recover astrophysically relevant elastic cross sections, but neither threshold opening nor a larger elastic reference cross section guarantees a dynamically relevant radiative rate.

The low-threshold extension also clarifies the scope of possible external constraints. In the present construction the mediator couples only inside the dark sector; direct-detection or supernova bounds cannot be applied without specifying a Standard-Model portal. Conversely, the very light masses in Table 2 may require a treatment of dark-sector cosmology, plasma screening, or collective effects beyond the isolated two-body kernel. We therefore use these points as controlled gravothermal tests, not as a complete cosmological model.

5.2. What B1938 does and does not imply

The numerical zero in Eq. (18) should not be interpreted as a detection of dSIDM or even as evidence for an ordinary cosmological NFW subhalo. It is obtained at the unevolved boundary with two fitted profile scales and two mass constraints. The isolated extrapolation is unusually concentrated. Even after profiling over a generous smooth truncation envelope, matching the two masses requires a large positive concentration offset and pushes the truncation to its allowed boundary. These findings weaken the interpretation of the zero-time solution as an ordinary cosmological NFW subhalo, but they are not a concentration measurement: a stripped-subhalo interpretation requires information not contained in the two masses. The explicit host-orbit sensitivity calculation shows that line-of-sight geometry can relieve the apparent tidal tension at the projected radius. It does so by reweighting the published one-dimensional pseudo-Jaffe summaries, however, and does not turn the initial NFW interpolation into a cosmological posterior. The evolved dissipative runs are also indistinguishable from the elastic controls. Within this analysis, B1938 therefore supplies no positive evidence for radiative self-interactions.

The result also does not exclude dSIDM in general. The observational input is compressed to two projected masses, the initial halo is restricted to an isolated NFW family, and the microphysical scan covers two long-range identical-fermion channels. A different emission normalization or channel, a nonperturbative scattering treatment, or a halo modified by tides and assembly history could yield a different cooling hierarchy. A full statistical claim would require production inference with the full visibility likelihood, the correlated perturber–macromodel posterior, and calibrated priors on halo mass, concentration, orbit, infall history, and formation time. The visibility framework and host-orbit sensitivity calculation developed here establish the data, likelihood, and prior components needed for that extension.

The present null result is nevertheless physically informative. It identifies a necessary condition for using B1938 as a dSIDM probe: after the candidate is placed in physical units, its relative-speed distribution must produce a cooling time comparable to the available evolution time. Source-frame threshold activation or a successful constant-model rescaling is insufficient.

5.3. Numerical scope

The final results use the spherical gravothermal approximation. This is appropriate for the controlled comparison of transport and local cooling in an isolated halo, but it does not include triaxiality, mergers, tidal stripping, or stochastic particle dynamics. Independent N-body validation would be required in a parameter region where cooling measurably changes the fluid trajectory. For the present candidates, however, the decisive hierarchy is the exact microscopic cooling time, which exceeds any astrophysical time by more than 20 orders of magnitude even when the threshold is open. An N-body calculation cannot turn this inactive source into an observable cooling signal.

The shell-resolution study also shows that coarse grids can alter the selected snapshot. The 48-shell results are not used for quantitative statements; screening is performed at 96 shells and the final time scan and elastic controls at 192 shells. A broader parameter search should retain this separation between screening and final resolution.

6. Conclusions

We have performed a self-consistent gravothermal test of Born-valid, velocity-dependent dSIDM in the compact strong-lens perturber JVAS B1938+666. The main conclusions are:

(i) The constant-parameter cooling law is recovered as a limit of the emitted-energy moment, while massive vector and scalar emission introduce a fixed microscopic velocity scale.

(ii) At nominal mass hierarchies, large phenomenological dSIDM cross sections are incompatible with a controlled Born expansion. Varying the hierarchy opens a Born-valid elastic window at lower dark-matter masses.

(iii) The 154-run final 192-shell scan spans threshold velocities from 1 to 500 $\mathrm{km\,s^{-1}}$. The reference cross section reaches $10\,\mathrm{cm^2\,g^{-1}}$ in the low-threshold grid. Continuous source-profile optimization removes the earlier discrete-scale selection artifact.

(iv) The initial NFW family interpolates the two masses exactly, as expected for two fitted scales. Its isolated extrapolation has $c_{200} \simeq 219$, 10.46 assumed-scatter units above the adopted median. A broad tidal-envelope profile still requires at least 6.36 units (5.69 for an aggressive bound) to reach the two-mass reference contour; these are sensitivity diagnostics, not formal exclusion significances.

(v) An independent audit of the GM068 visibility data reproduces the published data-volume bookkeeping and implements a normalized complex visibility likelihood with finite pixel-source marginalization. It provides an independent consistency check on the observational compression and anchors the data and likelihood layer of the analysis.

(vi) The published free-truncation pseudo-Jaffe summary implies a mean-parameter lower bound $r_{\mathrm{3D}} \geq 6.28$ kpc under the adopted tidal scaling, compared with the projected separation $R_{\mathrm{proj}} \simeq 1.52$ kpc. A phase-mixed host-orbit sensitivity prior gives a median current radius of 8.57 kpc. This demonstrates the importance of line-of-sight geometry but is not a formal orbit constraint or a discriminator among CDM, SIDM, and dSIDM.

(vii) The directly resimulated halos have $v_{\mathrm{max}} \simeq 5.34\,\mathrm{km\,s^{-1}}$. The 1 and 5 $\mathrm{km\,s^{-1}}$ branches are radiatively open, but the shortest cooling time is 8.7×10^{22} Gyr.

(viii) Cooling-disabled and dissipative direct runs give identical B1938 masses at fixed physical parameters for all open-threshold controls. The tested models are therefore in the effective elastic limit.

(ix) A 4018-point map over $v_\star/v_{\mathrm{max}}$ and σ_T/m_χ gives $\min(t_{\mathrm{cool}}/t_{\mathrm{dyn}}) = 8.3 \times 10^{26}$ for M1 and 1.2×10^{27} for M2. The null cooling conclusion is therefore not an artifact of the sparse direct grid or of testing only sub-threshold branches.

The B1938 mass measurements do not favor dissipative self-interactions in the explored Born-valid parameter region, but they do not exclude other dSIDM regimes. The GM068 audit and the host-orbit sensitivity calculation narrow the requirements for the next step, but they do not convert the two-mass compression into a model exclusion. A fully joint visibility-level reconstruction with adaptive source regularization, a correlated lens posterior, cosmological orbit and infall priors, and explicit evidence convergence is the natural next extension. A wider fluid or N-body scan of the same inactive kernels is not warranted; a different microscopic branch with a substantially larger validated emission moment remains physically distinct.

Appendix A. GM068 visibility-quality audit

Figure A1 records the reproducible quality-control stage used to construct the visibility likelihood. Panel (a) shows a representative positive-weight Stokes-I sample and its conjugate in the uv plane. Panel (b) displays adjacent-difference noise estimates in 30-minute baseline bins together with their time-bin median; these estimates also define the declared six-median-absolute-deviation RFI rule. Panel (c) compares the baseline-level empirical noise with the RMS inferred from the pipeline weights and exposes their arbitrary overall normalization. These diagnostics define the exclusions and internal noise scales used by the likelihood. A full-data dirty image additionally verifies the phase and sign conventions and identifies the global registration shift relative to the PRONTO coordinate system.

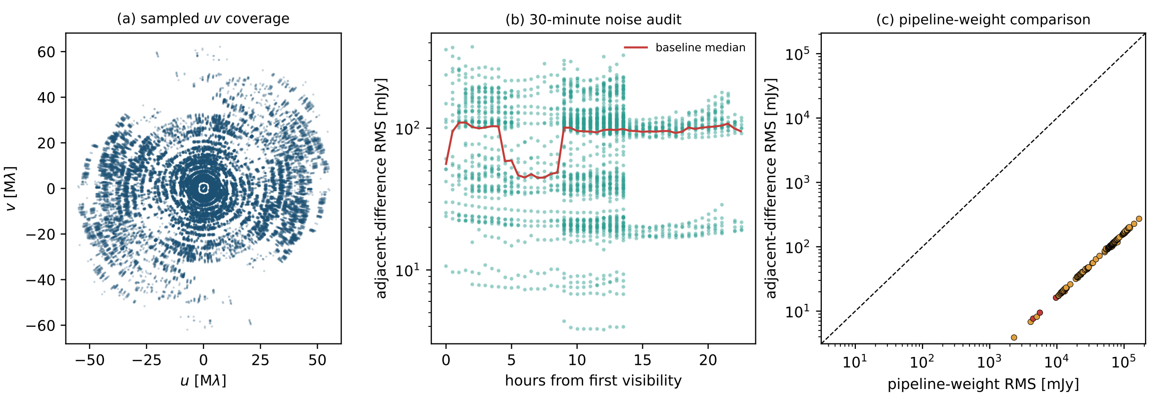

Figure A1. GM068 UVFITS quality audit: sampled uv coverage, 30-minute adjacent-difference noise estimates, and a baseline-level comparison with the pipeline-weight RMS. The AIPS weights and amplitudes have an arbitrary overall normalization, so the figure supports reproducibility of selection and internal noise weighting, not an absolute flux-calibration claim.

Code availability

The source code, simulation data, and figure-generation scripts underlying this work are publicly available at https://github.com/billkang-x/graverthermal-sidm.

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